Bangladesh University of Engineering and Technology



Department of Electrical and Electronic Engineering

A Project on

Resistor Color Code or Value Detector from Image

Course No: EEE 212

Course Title: Numerical Technique Laboratory

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Level: 2 Term: 1
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Objective:

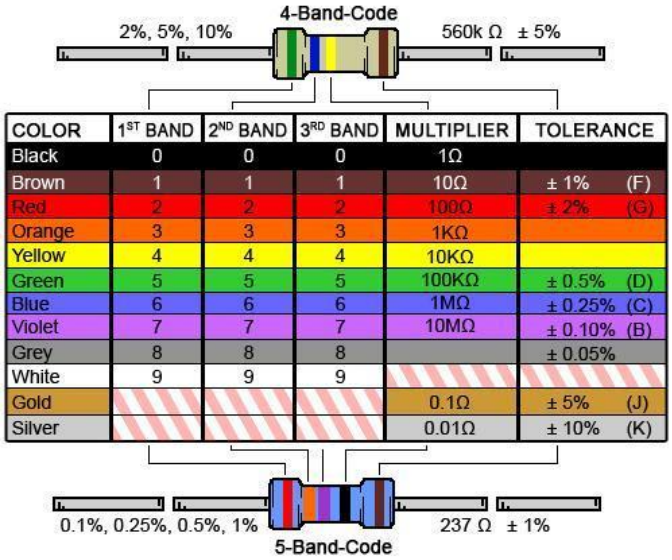
A resistor is an electrical component that implements resistance as a circuit element. So, identifying the value of resistance from the color codes of resistors is very important. Calculating the resistance manually is very time consuming; in case of complex circuits, it becomes even more troublesome.

So, we have used MATLAB for a faster solution for calculating values of multiple resistors, regardless of how they are placed. Our project emphasizes on taking resistor pictures as inputs and calculates the values of given resistors. Our purpose was to use MATLAB programs to detect the color bands of resistors, find out the sequence of the bands and calculate the resistance values using the color bands.

We have also built a GUI (Graphical User Interface) to make the program user friendly. Using this GUI, users can easily select the desired picture as input and the program will detect the resistors in the picture, calculate its color codes and show both the image and the value of all the resistors separately as output. Our objective for this project was to work on image processing and use image processing tools to detect the color bands of resistors and calculate the values for output without even needing to input the color codes manually for each resistor.

Theory:

Resistors have different color bands on the body which are used to identify their resistance values. Any resistor may have 4 or 5 color bands including a tolerance band. The code for each band and their sequence are used for calculating the resistance values. The following image illustrates the procedure of calculation:



COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

Fig-1: Color coding for 4 and 5 band resistors.

The tolerance band is often placed on one end and kept at a distance from the other color bands. This band is kept for adding or subtracting a percentage of error from the values calculated. The rest of the color bands are used for calculating the values. Each color of band has a specific number. The procedure of calculating the resistance value:

Resistance = Color code (band1), Color code (band2) $\times 10^{\text{color}}$ (Multiplier band) ± Color (Tolerance band) in Ohm (Ω)

For example, the following resistor has 4 bands: brown, black, silver and gold respectively.



If we put these values in the equation, the resistance will be

= Brown, Black x 10^{silver} ± Gold Ω

= 10 x 0.01 ± 5% Ω

= 0.1 ± 5% Ω

Challenges:

The major challenges that we faced while doing the project were:

- Evaluating values for multiple resistors, regardless of their orientation being parallel.
- Finding resistors amongst other electrical components such as capacitors, MOSFETs, inductors, LEDs, switches etc.
- Detecting the color bands of resistors even if the pictures are very bright or have shadows.
- Detecting the color bands even if the same color has different shades. Also calculating values of resistors with the same color for two or three bands.
- Detecting color bands in the desired order even if the orientations of the resistors are vertical, horizontal or angled.

Procedure:

- **Segmenting Images of the Resistors from Main Image:**

  The first job is to take an image as input from the user, the format can be any readable image format with RGB color space. Then the program detects the objects present on the image using various image segmentation techniques and creates a binary mask. From this binary mask, information about the objects can be extracted.

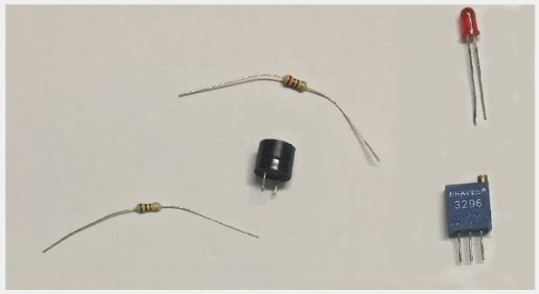


Fig-3: Main Image

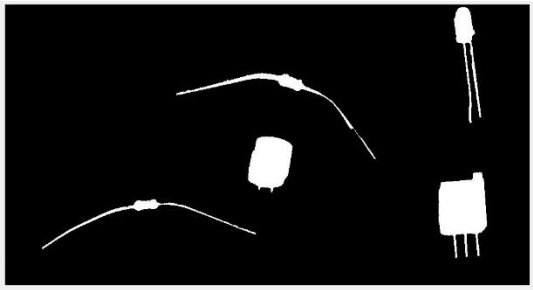


Fig-4: Binary Mask

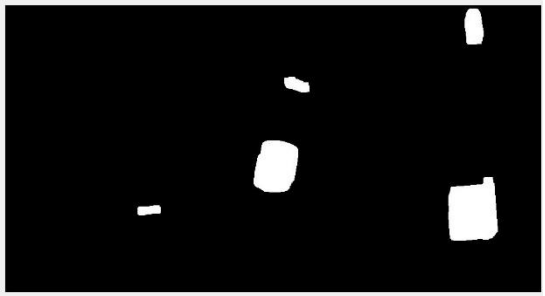


Fig-5: Morphological Structuring

- **Cropping and detecting the resistors from other components:**

  Using the binary mask, images of each component is cropped. Because every resistor has a unique light brown color, we can use this to differentiate from other electrical components. So, by using a mask, we detect the resistors and keep only the cropped images of resistors.

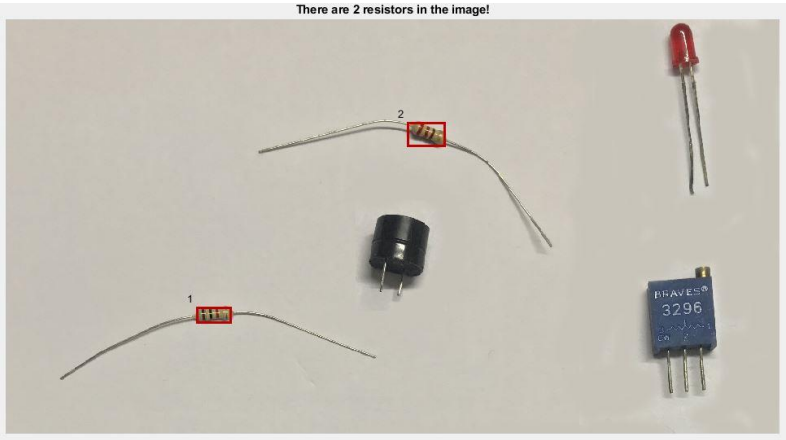


Fig-6: Detection of Resistors

- **Detecting the Color Bands along with values for each Resistor:**

  After we have cropped the images of the resistors, we can concentrate on detecting the color bands. For that, the program has different mask functions for different colors. When going through these functions, a structure array is created which includes information and values of different detected bands.

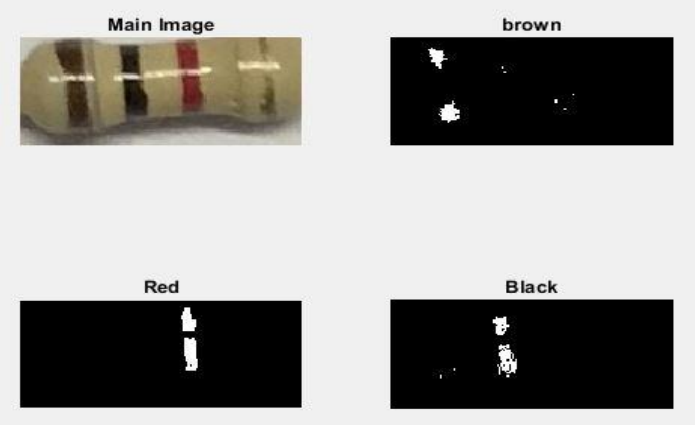


Fig-7: Binary Images from Different Color Masks

- **Calculating Value of the Resistor using Different Band Values:**

  After determining the corresponding color code values for Band1, Band2 and Band3, we calculate the value of resistance using,

  Resistance = (Band1, Band2) × 10^{Band3} ± Tolerance Band

- **Showing both Resistor Image and Value as Output:**

  The output will be shown according to how it was marked in the main image with the same numbers, both the value and image of the resistor will be shown.

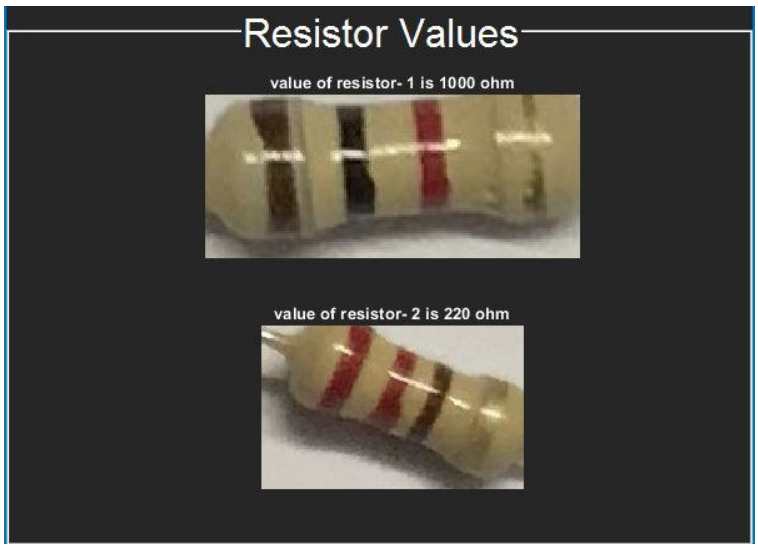
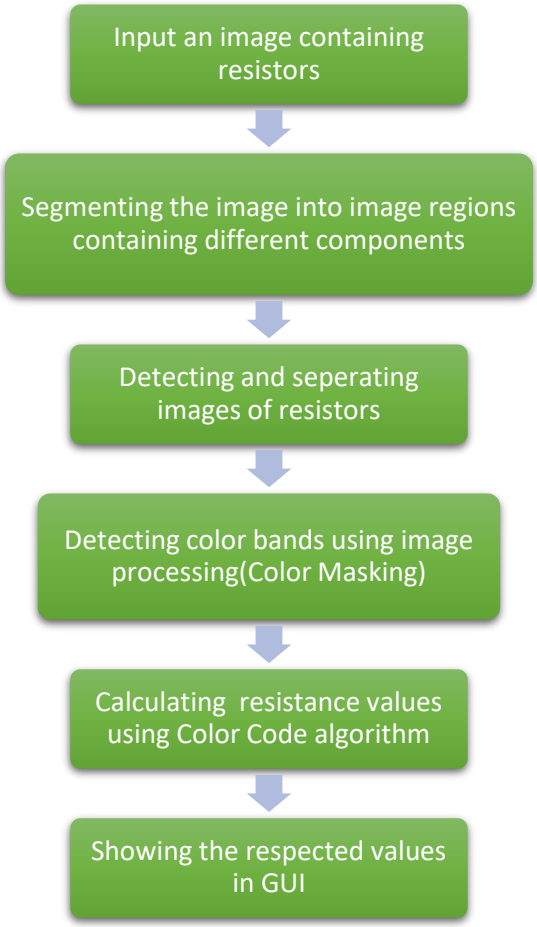


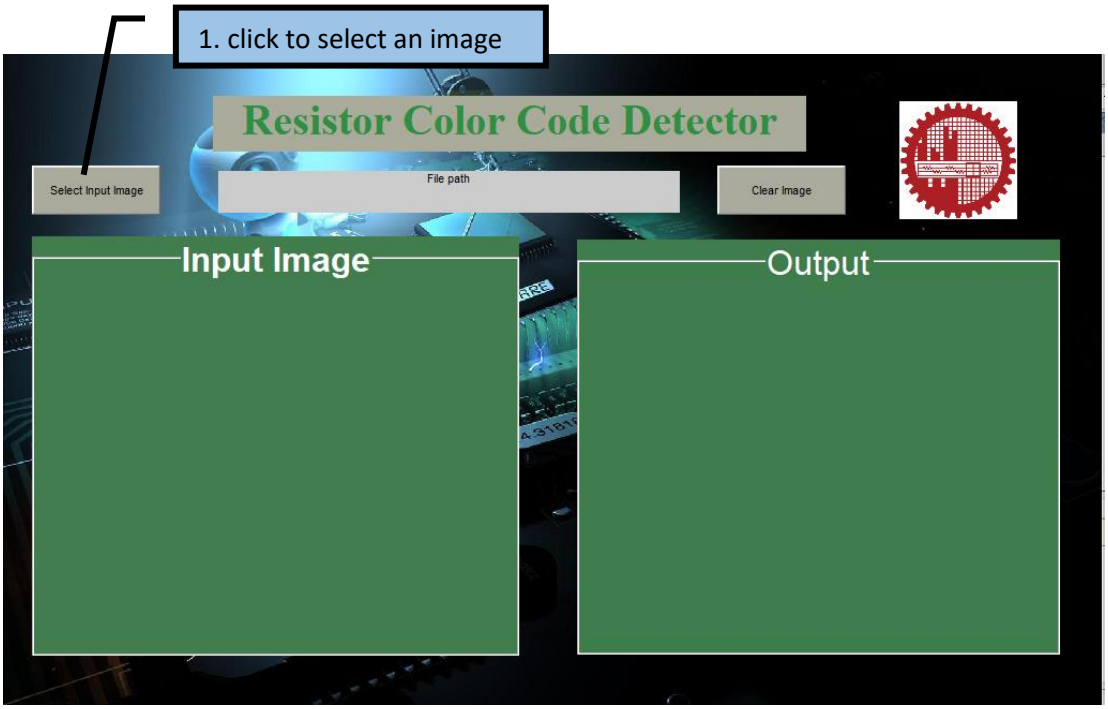
Fig-8: Final Output with Resistor Images and Values

Program Working Flowchart:

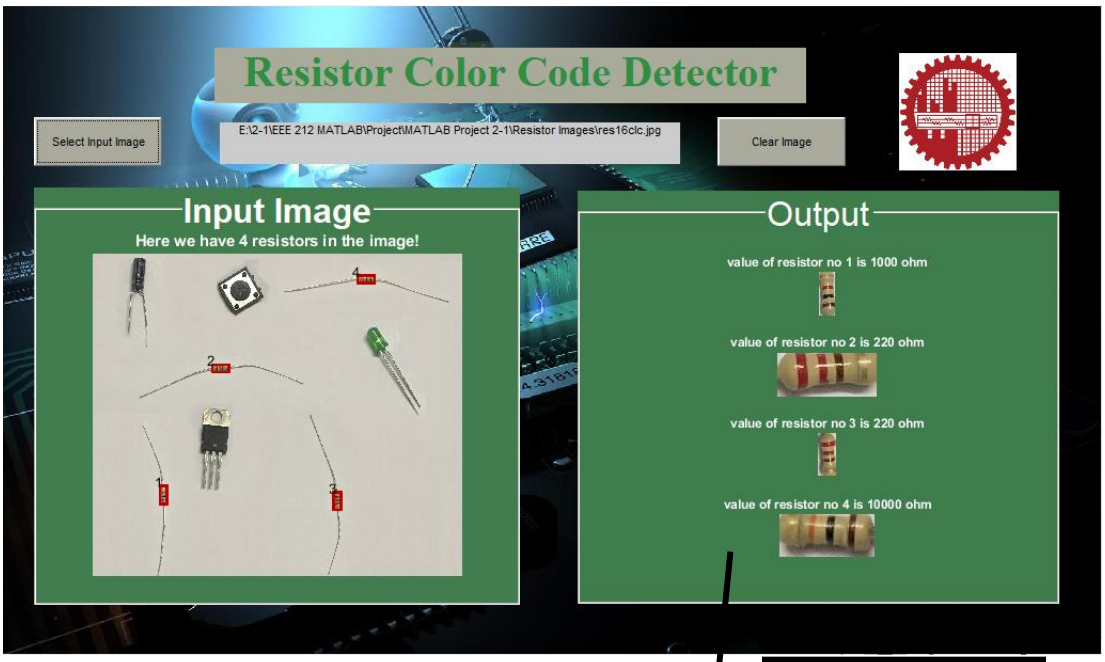


How to Use:

1. First, the user needs to run “Graphical_interface.m” in MATLAB. When the graphical user interface (GUI) comes up, the user needs to click on “Select Input Image”. This will open the file explorer for opening an image from anywhere of the hard drive of the computer. The process is illustrated below:



2. The program will start working. After it has finished analyzing the images, in the “Input Image” panel, the selected image will be shown with the detected resistors numbered and boxed out. In the “Output” panel, each resistor’s image will be shown separately along with their value in Ohm(Ω).



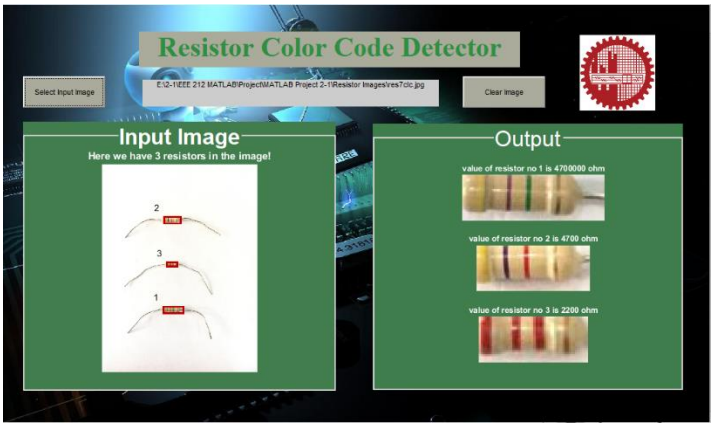
3. To select another image, user need to click on "Clear Image" first. Then the user can go on to select a new image following the process already discussed. It is mandatory to click on "Clear Image" first before selecting a new image.

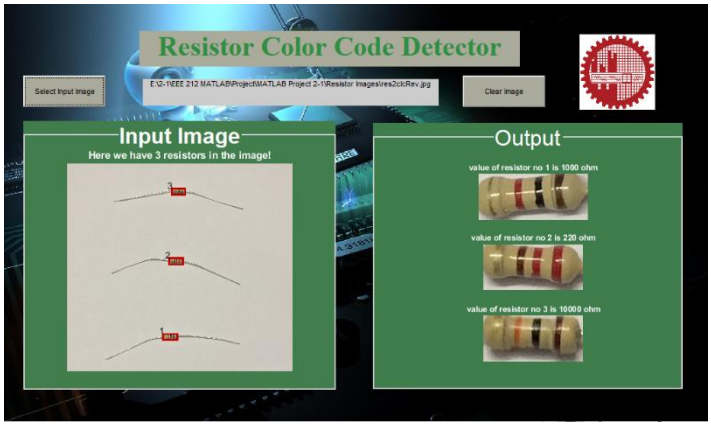


Features:

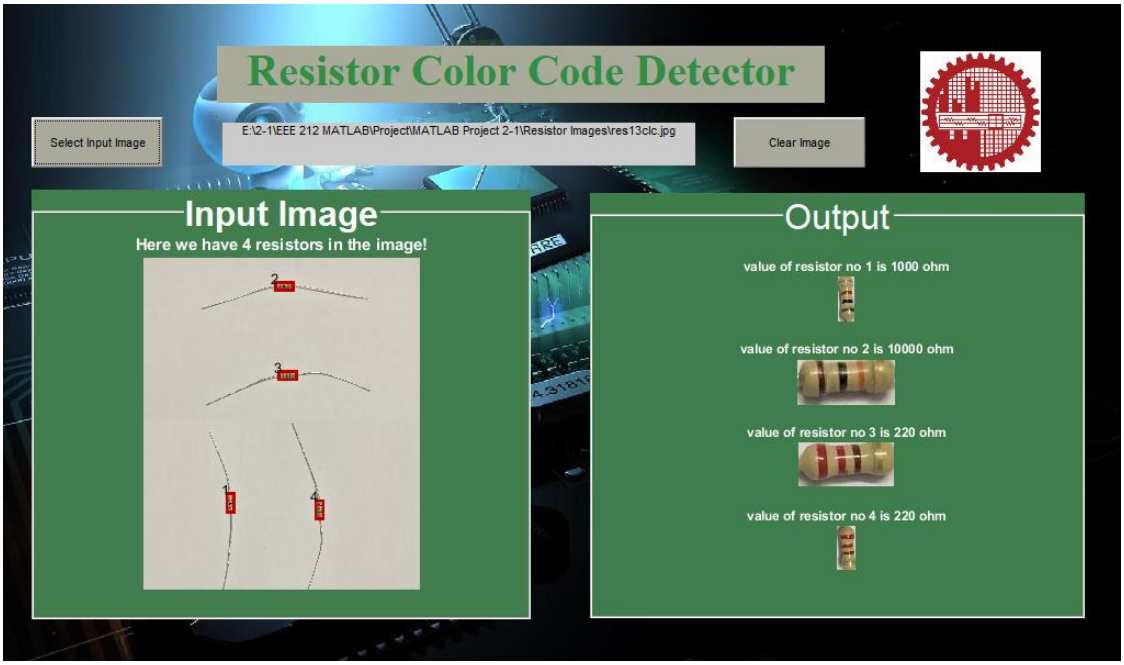
This program possesses some versatile features which make this user-friendly and precise. The main features are:

- Works on images with different level of brightness and shadow.



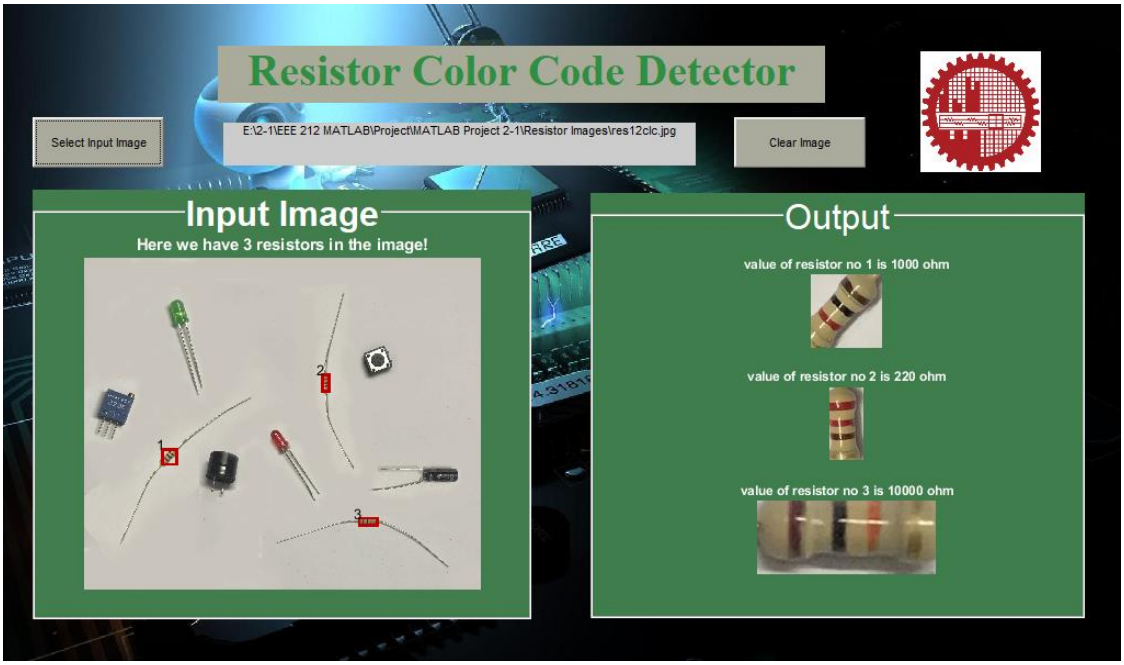


- Works with different orientation of resistor, it really doesn't matter how the resistors are placed, it will show the correct values.



- Can find the resistors precisely even if they are lying amongst different other electrical components like LEDs, MOSFETs, inductors, switches, buzzers etc.

The resistors are indicated with numbers so user can easily understand which value refers to which resistor.



Discussion:

The motive of our project was to detect and determine values of resistors from a picture input. The pictures contained electrical components other than multiple resistors. First, we detected the resistors and eliminated the other elements from the input pictures. Then we used those cropped resistor pictures and detected the color bands using image processing.

The pictures that we used were clicked using original resistors and circuit elements. We worked our way through the difficulties step by step and tried to solve them with efficiency. Though there might be a few limitations, our codes work perfectly for the inputs that we have provided. We also attempted to make a GUI so that the process becomes easier for the user. The GUI also works perfectly and shows accurate outputs.

Regardless of the difficulties, our attempts were quite satisfactory. With a bit more efficiency we might be able to simplify the code making it easier to understand. Even though the codes work perfectly, the total length is quite a lot. The codes and all the techniques that we used are easy to understand despite of the length. So, the overall project is user friendly and appropriate.